

SELECT & WHERE CLAUSE

SQL Tutorial In Hindi-5

Creating a Classroom dataset for practice

```
CREATE TABLE classroom
(  
    rollno int8 PRIMARY KEY,  
    name varchar(50) NOT NULL,  
    house char(12) NOT NULL,  
    grade char(1)  
);
```

```
INSERT INTO classroom  
(rollno, name, house, grade)  
VALUES  
(1, 'Sam', 'Akash', 'B'),  
(2, 'Ram', 'Agni', 'A'),  
(3, 'Shyam', 'Jal', 'B'),  
(4, 'Sundar', 'Agni', 'A'),  
(5, 'Ram', 'Yayu', 'B');
```

SELECT Statement

The SELECT statement is used to select data from a database.

- **Syntax**

```
SELECT column_name FROM table_name;
```

To select all the fields available in the table

- **Syntax**

```
SELECT * FROM table_name;
```

To select distinct/unique fields available in the table

- **Syntax**

```
SELECT DISTINCT Column_name FROM table_name;
```

WHERE Clause

The WHERE clause is used to filter records.

It is used to extract only those records that fulfill a specified condition

- **Syntax**

```
SELECT column_name FROM table_name  
WHERE conditions;
```

- **Example**

```
SELECT name FROM classroom  
WHERE grade='A';
```

Operators In SQL

The SQL reserved words and characters are called operators, which are used with a WHERE clause in a SQL query

Most used operators:

- 1. Arithmetic operators :** arithmetic operations on numeric values
Example: Addition (+), Subtraction (-), Multiplication (*), Division (/), Modulus (%)
- 2. Comparison operators:** compare two different data of SQL table
 - Example: Equal (=), Not Equal (!=), Greater Than (>), Greater Than Equals to (>=)
- 3. Logical operators:** perform the Boolean operations
 - Example: ALL, IN, BETWEEN, LIKE, AND, OR, NOT, ANY
- 4. Bitwise operators:** perform the bit operations on the Integer values
 - Example: Bitwise AND (&), Bitwise OR(|)

LIMIT Clause

The **LIMIT** clause is used to set an upper limit on the number of tuples returned by SQL.

Example: below code will return 5 rows of data

```
SELECT column_name FROM table_name  
LIMIT 5;
```

ORDER BY Clause

The **ORDER BY** is used to sort the result-set in ascending (ASC) or descending order (DESC).

Example: below code will sort the output data by column name in ascending order

```
SELECT column_name FROM table_name  
ORDER BY column_name e ASC;
```


IMPORT CSV FILE

SQL Tutorial In Hindi-6

CSV files

customer csv file: <https://bit.ly/3KLPb2P>

payment csv file: <https://bit.ly/41kIYjW>

STRING FUNCTION

SQL Tutorial In Hindi-7

Functions In SQL

Functions in SQL are the database objects that contains a set of SQL statements to perform a specific task. A function accepts input parameters, perform actions, and then return the result.

Types of Function:

1. System Defined Function : these are built-in functions
 - Example: rand(), round(), upper(), lower(), count(), sum(), avg(), max(), etc
2. User-Defined Function : Once you define a function, you can call it in the same way as the built-in functions

Most Used String Functions

String functions are used to perform an operation on input string and return an output string

- **UPPER()** converts the value of a field to uppercase
- **LOWER()** converts the value of a field to lowercase
- **LENGTH()** returns the length of the value in a text field
- **SUBSTRING()** extracts a substring from a string
- **NOW()** returns the current system date and time
- **FORMAT()** used to set the format of a field
- **CONCAT()** adds two or more strings together
- **REPLACE()** Replaces all occurrences of a substring within a string, with a new substring
- **TRIM()** removes leading and trailing spaces (or other specified characters) from a string